Bank Loan Performance & Risk Analysis

Data-Driven Insights using SQL & Tableau

INTERNSHIP

Prepared by

Anushka Sonawane

Offer Letter ID

APSPL2628018

Tools & Technologies

SQL | PYTHON | TABLEAU

Problem Statement

Analyzing bank loan data to evaluate performance and identify critical risk factors.

Core Objective

Leveraging SQL and Python to transform raw loan data into actionable financial insights that support data-driven decision-making.

Trend Analysis

Analyze total loan applications and funding trends to monitor growth.

Loan Health

Identify Good vs Bad Loans to assess portfolio quality and loss risk.

Regional Performance

Evaluate monthly and regional performance to pinpoint high-growth areas.

Customer Behavior

Understand behavioral patterns including loan purpose, term, and employment.

Strategic Outcome

Drive long-term profitability by identifying risk factors early and optimizing loan approval processes.

Dataset Overview

The dataset includes detailed loan-level information for comprehensive financial analysis.

Key Fields: Financial

- Loan Amount, Total Payment
- Interest Rate, DTI
- Loan Status (Fully Paid, Current, Charged Off)

Key Fields: Contextual

- Issue Date (Monthly trends)
- State & Geographic distribution
- Purpose, Term, Home Ownership

Analysis Approach

A dual-tooled methodology for precise extraction and visual storytelling.

SQL
TABLEAU

Data Depth

Leveraging high-granularity loan-level data to evaluate performance metrics and identify critical risk factors across multiple dimensions.

Key KPIs

Volume Metrics

Total Loan Applications
`COUNT(id)`

Total Funded Amount
`SUM(loan_amount)`

Total Amount Received
`SUM(total_payment)`

Performance Metrics

Good Loan %
(Fully Paid + Current)

Bad Loan %
(Charged Off)

Risk Metrics

Average Interest Rate
Mean rate across portfolio

Average DTI
Debt-to-Income Ratio

Time-Based Analysis (MTD vs PMTD)

Strategic Insight

Comparing Month-to-Date (MTD) against Previous Month-to-Date (PMTD) enables the identification of critical growth trends and seasonal patterns within the loan portfolio.

Tracking Performance Metrics

Loan Volume

Loan Applications
MTD vs PMTD

Funding

Funded Amount
MTD vs PMTD

Revenue

Amount Received
MTD vs PMTD

Risk Factors

Interest Rate & DTI
MTD vs PMTD

Execution: SQL queries are utilized to extract precise comparative data points for these metrics across time periods.

Loan Status Analysis

Status Categories

Fully Paid **Current** **Charged Off**

Standard classifications used for portfolio tracking and risk assessment.

Key Observations

Majority loans are Good Loans

Good loans contribute highest revenue recovery

Risk Segment Identified

Charged Off loans represent a critical risk segment

Monthly Trend Analysis

Technical Implementation

```
(SQL: GROUP BY MONTH(issue_date))
```

Aggregating records by issuance month to reveal temporal patterns.

Key Insights

Application Variance

Loan applications vary across months, identifying peaks in borrower activity.

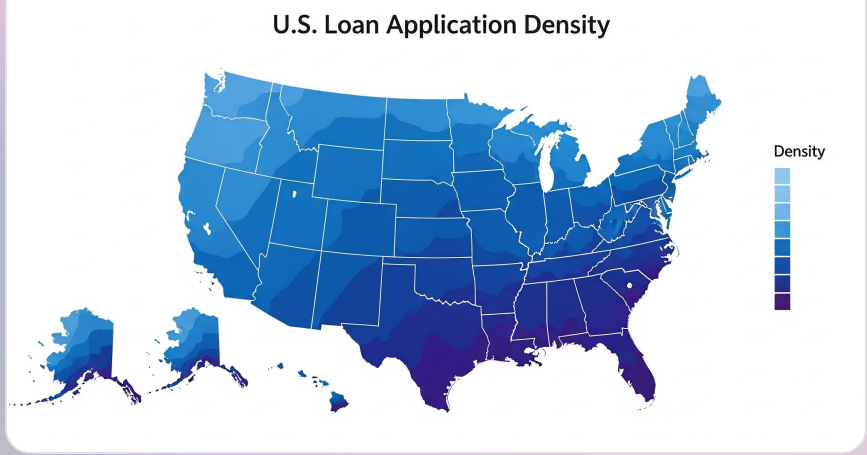
Seasonal Demand

Certain months show higher loan demand, helping optimize resource allocation.

Future Forecasting

Historical data helps in forecasting future loan trends and portfolio growth.

State-wise Analysis



Technical Implementation

`(SQL: GROUP BY address_state)`

Segmenting the customer base by state to identify regional trends.

Key Insights

Volume Distribution

Some states have higher loan applications, indicating regional concentration of customers.

Strategic Value

Data is useful for developing targeted marketing strategies for high-volume regions.

Customer Behavior Analysis

Analysis Dimensions

Purpose • Term • Employment • Home Ownership

Multi-dimensional filtering allows for deep dives into specific borrower archetypes.

Key Insights

Market Dominance

Certain purposes, such as debt consolidation, significantly dominate customer borrowing behavior.

Segmented Demand

Loan demand varies noticeably across different employment categories and professional backgrounds.

Strategic Value

Analyzing these behaviors helps the organization better understand and serve specific customer financial needs.

Risk Analysis

Analysis Variables

`Loan Status • DTI • Payment Behavior`

These key metrics are used to evaluate borrower stability and predict creditworthiness.

Key Risk Insights

Debt-to-Income (DTI) Impact

Higher DTI ratios directly correlate to a higher risk of default.

Credit Risk Segments

Charged Off loans serve as primary indicators for identifying high-risk segments.

Loan Purpose Variance

Specific loan purposes exhibit inherently higher levels of risk than others.

Dashboard Overview

Features

`Filters • KPI Cards • Monthly Trends • Loan Status`

Includes interactive filters for State, Purpose, and Grade, alongside comprehensive trend breakdowns.

Key Benefits

Easy Exploration of Data

Intuitive navigation and interactive elements allow for deep dives into borrower datasets.

Real-time Decision Support

Empower leadership with up-to-the-minute metrics to inform lending strategies immediately.

Visual Understanding of Trends

Transform complex numbers into clear visual narratives to identify growth patterns easily.

Strategic Recommendations

Focus on High-Performing States
Prioritize resource allocation to regions demonstrating robust lending metrics.

Improve Risk Assessment
Enhance screening protocols specifically for high DTI customer segments.

Reduce Charged Off Loans
Implement stricter screening to minimize defaults and protect the portfolio.

Promote High-Demand Categories
Scale marketing efforts for loan products with high market traction.

Data-Driven Strategic Planning
Leverage advanced analytics and insights for long-term organizational goals.

Conclusion

Portfolio Performance

Majority of loans are performing well, indicating a stable lending environment.

Balanced Outlook

Data reveals both opportunities and risks within the current market landscape.

Tech Integration

SQL and Tableau integration enables powerful, actionable insights for decision makers.

Full Lifecycle Capability

This project demonstrates end-to-end data analysis capability and professional rigor.